Multi-Agent Coordination Patterns

for Data-Driven Construction Automation

Based on: Anthropic Coordination Guidelines · DDC Skills Framework · Boiko (2025)

Construction AI Agent Swarm — Phase 5 Target Architecture

Agenda

01 **The Construction Data Problem**
Why construction lags behind other industries

02 **Five Coordination Patterns**
Definitions, selection criteria, trade-offs

03 **Construction Applications**
Cost estimation, schedules, BIM, documents

04 **DDC Agent Swarm Architecture**
Recommended production system design

Construction’s Data Problem

- **Last of 19 industries** in IT spending
- Banks decide in **minutes** — data only
- Agriculture decides in **hours** — data + experience
- Construction decides in **days** — opinion
- BIM and PMIS offer **highest ROI** potential
- Adoption remains **critically low**

> "The digital future of construction is not just about new technologies — it is fundamentally rethinking data handling and business models."
>
> — Boiko, DDC Book (2025)

Banking
Decisions in minutes — 100% data-driven

Agriculture
Decisions in hours — data + experience

Construction
Decisions in days — opinion-based

The DDC Framework: BIM + AI + Multi-Agent

BIM = Database

IFC files expose structured volumes, areas, materials via ifcopenshell — no 3D viewer required

55,719 CWICR Items

Standardized work item database in 9 languages — semantic search via Qdrant vector DB

Phase 5 Goal

Construction AI Agent Swarm: PLANNER · ESTIMATOR · SCHEDULER · VALIDATOR · REPORTER

Gartner 2025: Multi-agent system inquiries surged 1,445% from Q1 2024 to Q2 2025. 40% of enterprise applications will embed AI agents by end of 2026.

Five Coordination Patterns

Selecting the right architecture for each construction task

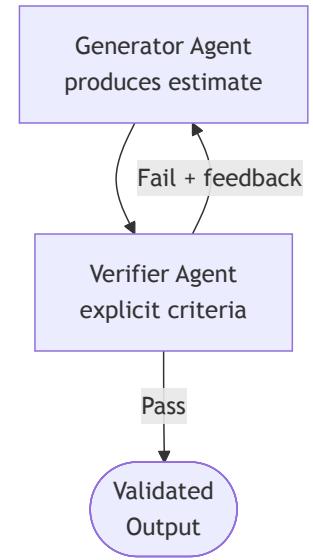
The Five Patterns at a Glance

Pattern	Primary Strength	Construction Application
Generator-Verifier	Quality-critical output, explicit criteria	Cost estimate validation, BIM IDS compliance
Orchestrator-Subagent	Clear decomposition, parallel subtasks	IFC → cost estimate pipeline
Agent Teams	Long-running specialist context	Multi-package estimating, multi-building
Message Bus	Event-driven, extensible ecosystem	RFI processing, document routing
Shared-State	Collaborative knowledge accumulation	Project knowledge base, benchmarks

Pattern 1: Generator-Verifier

Best for: Quality-critical outputs with explicit criteria

- Generator agent produces initial output
- Verifier checks against defined criteria
- Failure returns structured feedback to generator
- Loop continues until criteria are satisfied



Construction use cases

Cost estimate validation · BIM IDS compliance · Schedule logic checks

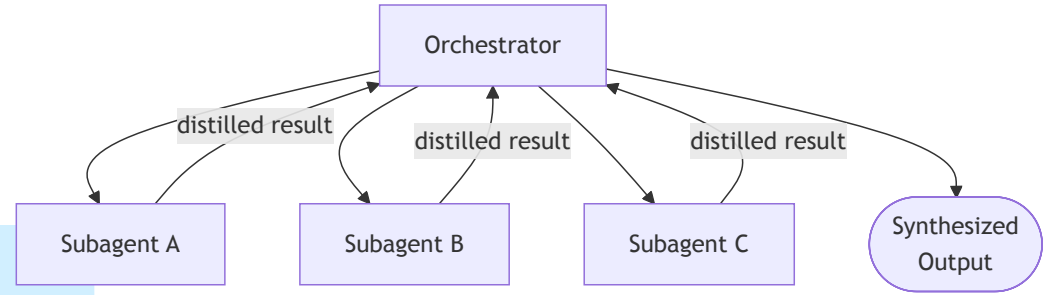
Key risk: early victory problem

Vague criteria produce rubber-stamp approval — be explicit

Pattern 2: Orchestrator-Subagent

Best for: Multi-step tasks with clear decomposition

- Orchestrator decomposes and delegates tasks
- Subagents work in isolated context windows
- Each returns distilled, compact findings
- Orchestrator synthesizes final output



Construction use cases

IFC → cost estimation · BIM analysis · Complex document processing

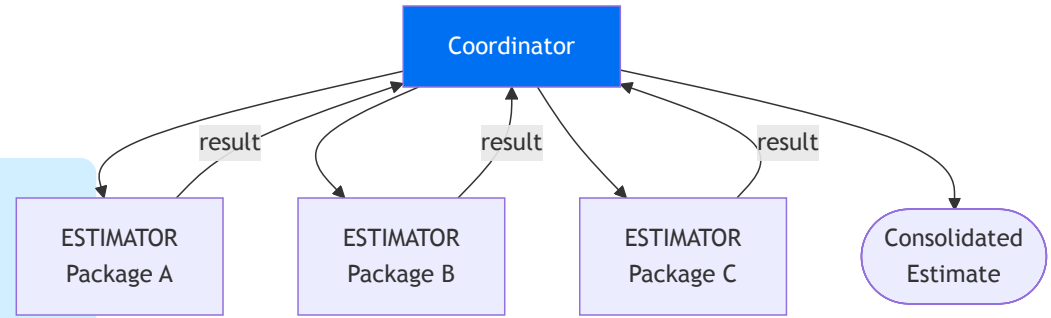
Key benefit: context protection

5,000+ token IFC data never enters the Rate Calculator context

Pattern 3: Agent Teams

Best for: Parallel long-running specialist analyses

- Coordinator assigns work from a shared queue
- Worker agents persist across many assignments
- Workers accumulate deep domain context
- Best for independent, partitioned workloads



Construction use cases

Multi-package estimating · Multi-building scheduling · Trade-specific analysis

Key requirement: independence

Teammates cannot share context directly — partition by trade or zone

Pattern 4: Message Bus

Best for: Event-driven, growing agent ecosystems

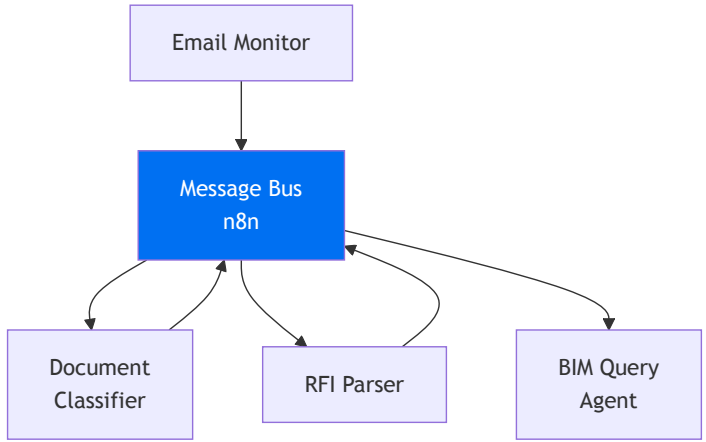
- Agents publish to topics and subscribe to events
- Router delivers matching messages automatically
- New agents added without rewiring existing ones
- Every event logged with timestamp and agent ID

Construction use cases

RFI processing · Change order tracking · Document routing · Submittal review

Key benefit: extensibility

New document types → new subscriber agents, no rewiring



Pattern 5: Shared-State

Best for: Collaborative knowledge accumulation

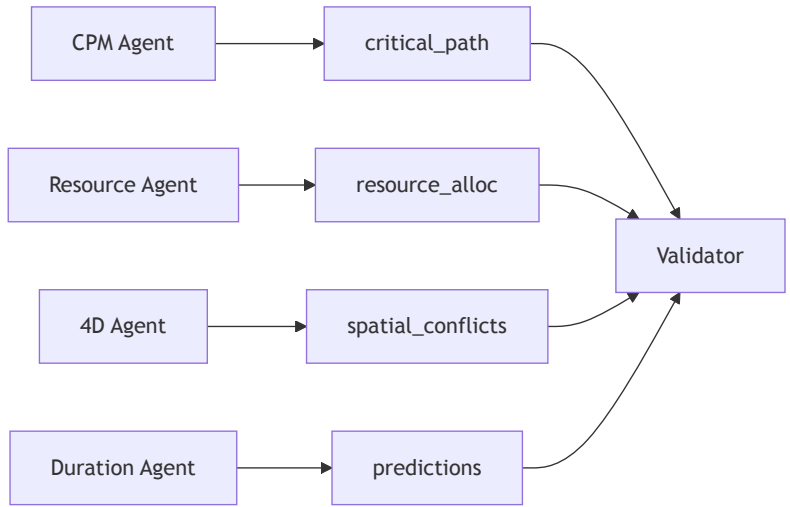
- No central coordinator — agents act autonomously
- All agents read and write to persistent store
- Write domains partitioned to prevent conflicts
- Termination: convergence signal from Validator

Construction use cases

Project knowledge base · Historical cost database · Cross-session institutional memory

Key risk: reactive loops

Each agent must write only to its designated domain



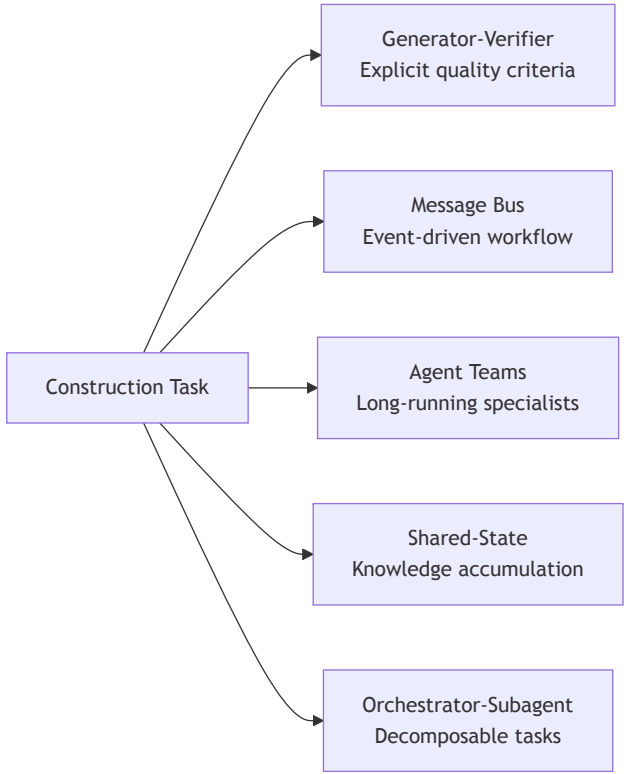
Pattern Selection Guide

Choose based on task characteristics:

- Explicit quality criteria → **Generator-Verifier**
- Event-driven, growing ecosystem → **Message Bus**
- Long-running parallel specialists → **Agent Teams**
- Knowledge accumulation across agents → **Shared-State**
- Multi-step decomposition → **Orchestrator-Subagent**

Production systems combine patterns

Primary: Orchestrator-Subagent + Shared-State knowledge layer + Generator-Verifier quality gate



Construction Applications

Deep dives: Cost Estimation · Scheduling · Documents · BIM

Recommended Hybrid Architecture

Orchestrator-Subagent

Primary workflow: decomposes tasks, dispatches specialized subagents, synthesizes outputs

IFC Parser · CWICR Matcher · Rate Calculator · Document Processor

Shared-State

Persistent knowledge layer: Qdrant + PostgreSQL accumulate institutional project knowledge across sessions

CWICR embeddings · Historical estimates · Document extracts

Generator-Verifier

Quality gate at every output: explicit criteria prevent false positives from reaching users

Estimate validation · Schedule verification · BIM compliance

Anthropic: "A common hybrid uses orchestrator-subagent for the overall workflow with shared state for a collaboration-heavy subtask."

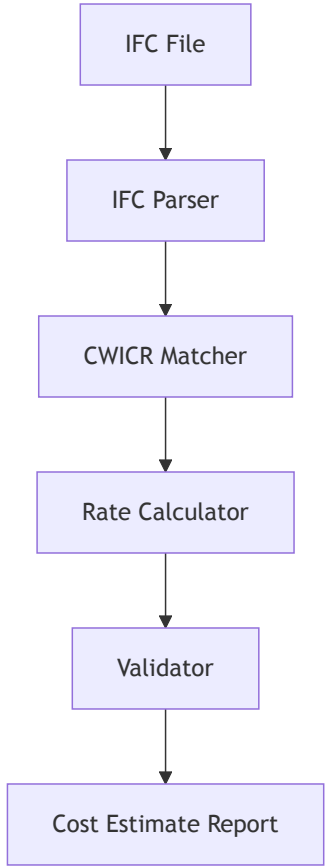
Cost Estimation Pipeline

16 hours → 2 hours (87% faster)

- **Stage 1:** IFC Parser extracts element inventory
- **Stage 2:** CWICR Matcher — Qdrant semantic search
- **Stage 3:** Rate Calculator — PostgreSQL unit rates
- **Stage 4:** Validator — statistical range checks
- **Stage 5:** Orchestrator synthesizes final report

Context isolation benefit

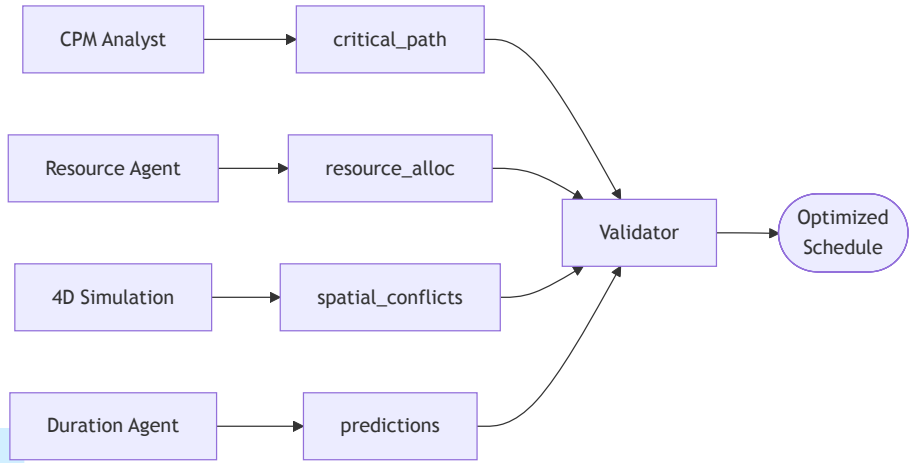
5,000+ token IFC element data never enters the Rate Calculator context — clean subagent boundaries



Schedule Optimization

Agent Teams + Shared-State

- **CPM Agent** writes critical path activities
- **Resource Agent** reads CPM, writes leveled durations
- **4D Simulation** reads both, writes spatial conflicts
- **Duration Agent** writes ML-predicted durations
- **Validator** monitors all sections — signals convergence



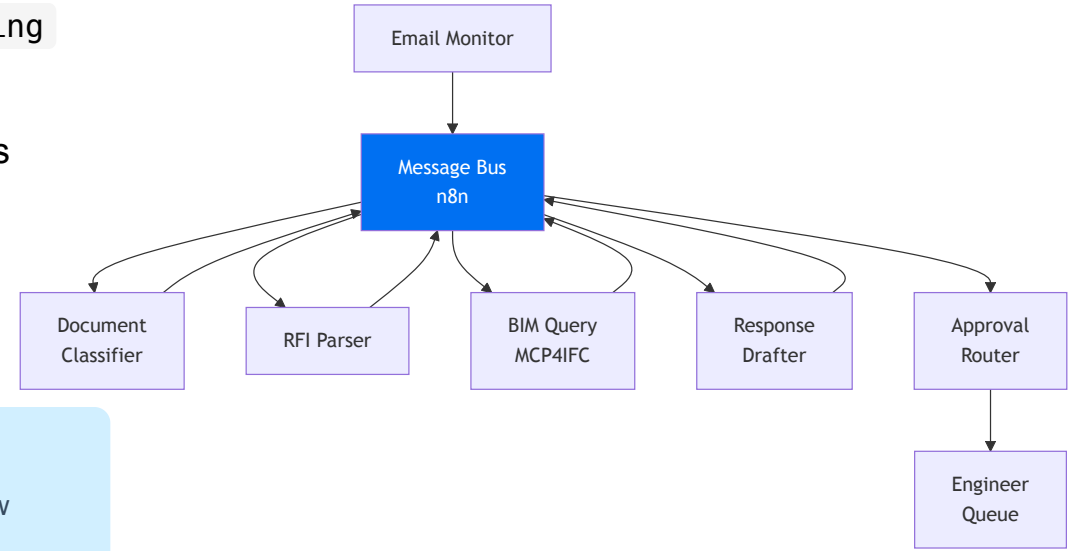
Write domain partitioning

Each agent writes only to its section — prevents reactive loops.
polars: 7-10x faster than pandas for large activity networks.

RFI Document Processing

Message Bus Pipeline

- **Email Monitor** publishes `documents.incoming`
- **Classifier** routes to `documents.rfi`
- **RFI Parser** extracts question and drawing refs
- **BIM Query** looks up elements via MCP4IFC
- **Response Drafter** generates draft answer
- **Approval Router** routes to engineer queue



Extensibility

Add submittal or change order agents by subscribing to new topics — no rewiring of existing agents

BIM Analysis Orchestration

Five-Subagent Pipeline

- **Subagent 1:** IFC Parser — spatial boundary filter
- **Subagent 2:** Element Classifier — OmniClass/UniClass
- **Subagent 3:** CWICR Matcher — top-5 similarity search
- **Subagent 4:** Rate Calculator — regional PostgreSQL rates
- **Subagent 5:** Validator — range checks + completeness

Provenance chain

Every estimate line traced to its IFC element — impossible in manual estimating. Full audit trail for dispute resolution.

1. IFC Parser
ifcopenshell · spatial boundary filter

↓

2. Element Classifier
OmniClass · UniClass · trade grouping

↓

3. CWICR Matcher
multilingual embeddings · Qdrant top-5

↓

4. Rate Calculator
regional PostgreSQL unit rates

↓

5. Validator
range + completeness checks · fail loops back

↓

Estimate Report with Provenance Chain

Technical Infrastructure

MCP Integration · Czech NLP · Open-Source Stack

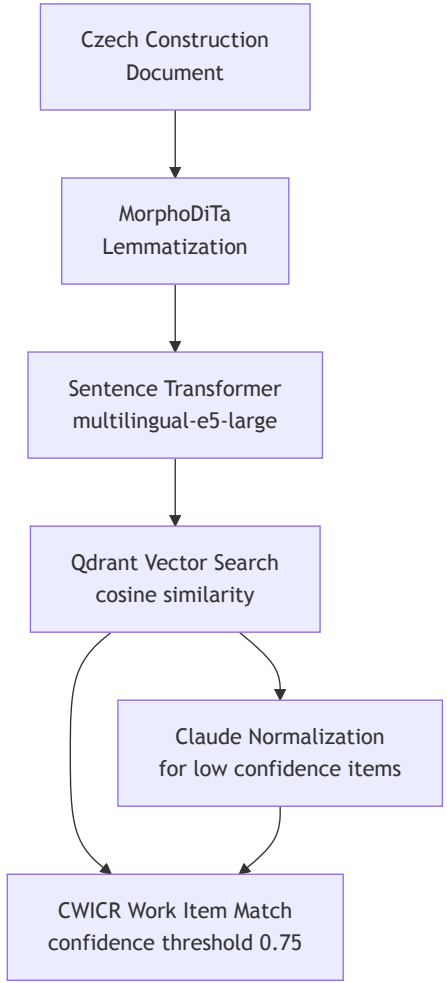
Czech Language Challenge

Morphological Complexity

- **7 grammatical cases** — one concept, 5+ surface forms
- "betonová zeď" → genitive, dative, instrumental, plural forms
- Standard regex misses **4 of 5** occurrences
- Construction CWICR covers **9 languages**

Three-layer solution

- MorphoDiTa lemmatization (Charles University UFAL)
- multilingual-e5-large sentence embeddings
- Qdrant vector search — confidence threshold 0.75



MCP Integration Architecture

Construction Agents

Orchestrator (Tool Search)

IFC Parser Subagent

CWICR Matcher Subagent

Rate Calculator Subagent

Document Processor Subagent

MCP Server Layer

Qdrant-MCP — search_work_items

PostgreSQL-MCP — rate tables

n8n-MCP — workflow triggers

Speckle-MCP — BIM data exchange

IFC Server MCP — 50+ BIM tools

External Data Systems

Qdrant DB (CWICR 55K items)

PostgreSQL (ERP + Rates)

n8n Workflows (400+ integrations)

Speckle Hub (BIM Models)

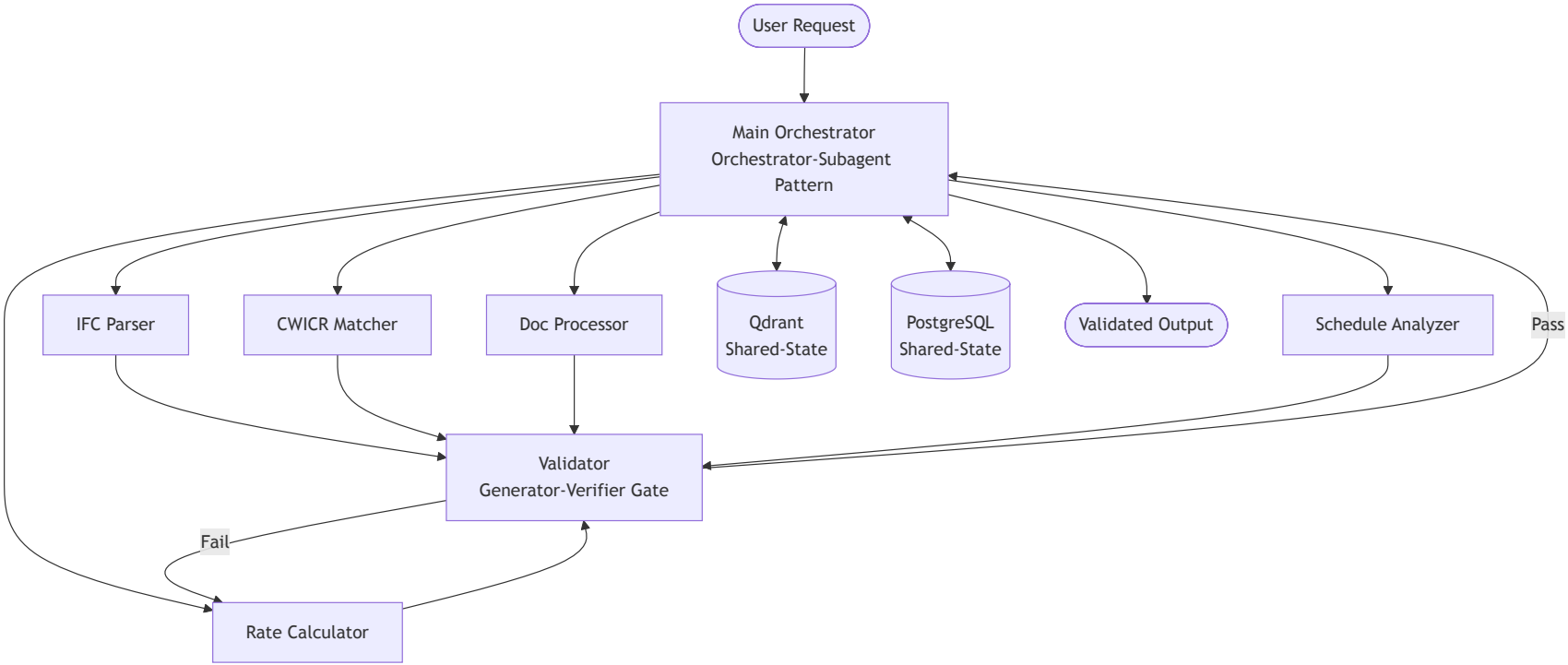
IFC OpenBIM Server

Tool Search: Reduces tool-definition tokens by 85% — loads only tools relevant to the current request, preventing context flooding

Open-Source Technology Stack

Category	Primary Tool	Why Choose It
BIM / IFC Processing	ifcopenshell + MCP4IFC	50+ LLM-native BIM tools, no license needed
Vector Search	Qdrant	100M+ vectors, rich payload filtering, Rust speed
Document Processing	Docling (IBM) + pdfplumber	Complex PDF layouts, table extraction
Data Processing	polars (7-10x faster than pandas)	Columnar, multi-threaded, lazy evaluation
Workflow Automation	n8n	400+ integrations, native AI agent nodes
BIM Collaboration	Speckle	Open-source, no Revit license, ACC integration

DDC Construction Agent Swarm



Conclusion

Key Takeaways

Five Patterns, Five Problems

Each pattern solves a distinct construction challenge — choose based on task characteristics, not convention

87% Time Saving Verified

Cost estimation: 16 hours → 2 hours via IFC → CWICR Qdrant → PostgreSQL rates pipeline

Hybrid is the Target State

Orchestrator-Subagent + Shared-State + Generator-Verifier is the recommended production architecture

MCP + Skills = Production

Skills provide procedural knowledge; MCP servers provide tool access — together they reach production construction systems

The agentic future of construction is already under construction in the DDC Skills repository

Thank You

Data-Driven Construction (DDC)

DDC Book (2025) — Artem Boiko

DDC Skills Repository — github.com/datadrivenconstruction

MCP4IFC Framework — Open-source BIM tools for LLMs

Anthropic — Multi-Agent Patterns Documentation

Complete Open-Source Stack

ifcopenshell · Qdrant · Docling · polars · n8n · Speckle · MorphoDiTa · scikit-learn